

Likelihood-free phylogenetics



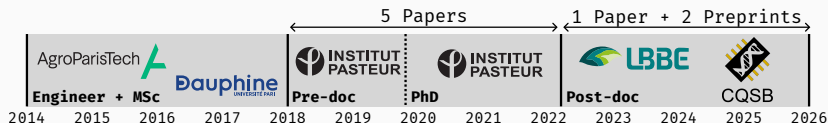
Luc Blassel

Auditions CNRS 51/02

May 20th 2026

slides: lucblassel.com/CNRS_2026.pdf

Personal research context & history



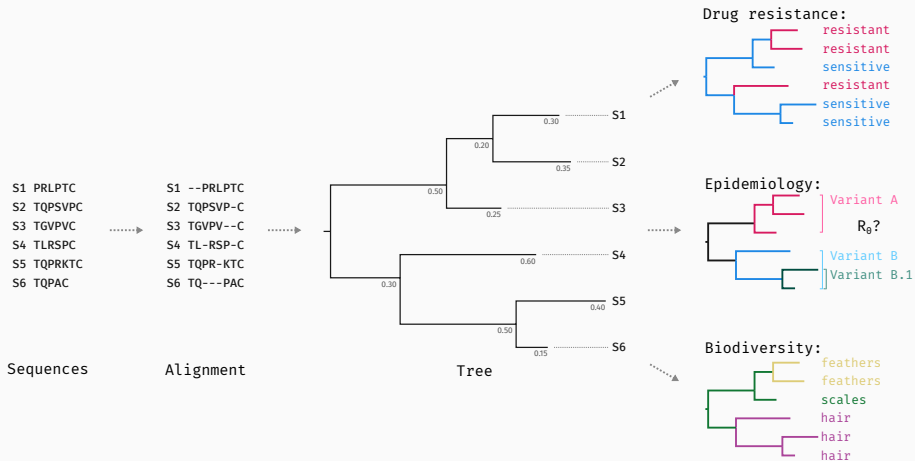
An interdisciplinary setting:

- *Algorithmics & theory*: **Improving** sequence **alignments**
- *Data analysis*: **Learning** from alignments
- *Methods & engineering*: **Deep Learning** for phylogenetic **inference**

General research framework:

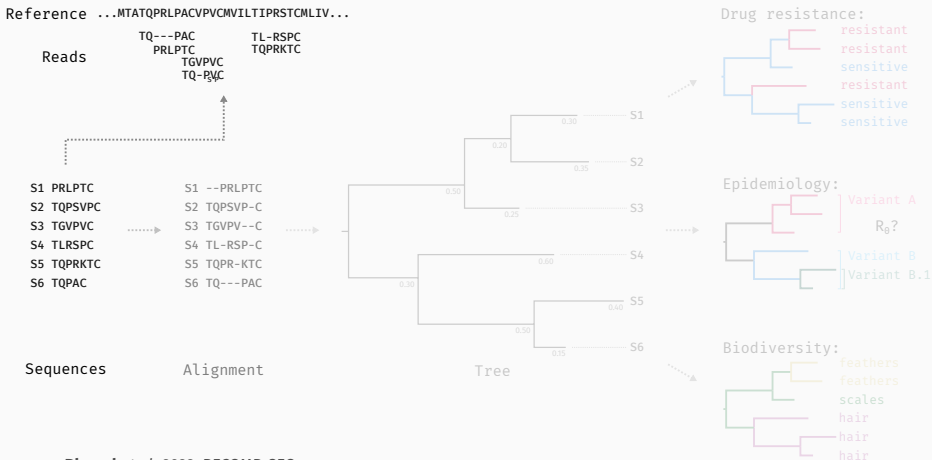
Methods to gather **evolutionary** insight from **sequences**

General research context - Phylogenetic analysis



Previous research work

keywords: Sequence Bioinformatics, combinatorics



Blassel et al. 2022, RECOMB-SEQ;

keywords: Sequence Bioinformatics, combinatorics

Reference ...MTATQPRLPACVPVCMVILTIPRSTCMLIV...

Reads
TQ---PAC
PRLPTC
TGVPVC
TQ-PVC
TL-RSPC
TQPRKTC

S1 PRLPTC
S2 TQPSVPC
S3 TGVPVC
S4 TLRSPC
S5 TQPRKTC
S6 TQPAC

S1 --PRLPTC
S2 TQPSVP-C
S3 TGVPV--C
S4 TL-RSP-C
S5 TQPR-KTC
S6 TQ---PAC

Sequences

Alignment

Summary:

Goal: improve read-mapping

Method: sequence-to-sequence transformations

Impact: Adapted in a genome assembly tool ¹

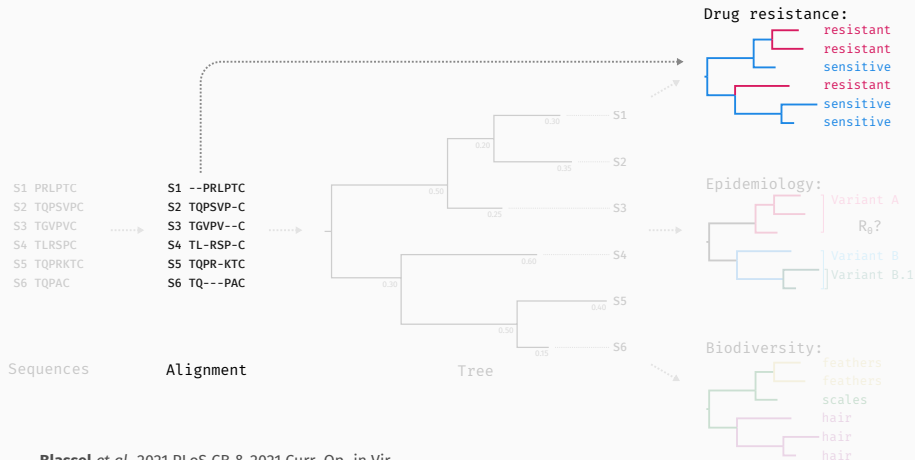
Drug resistance:
resistant

Tree

Biodiversity:

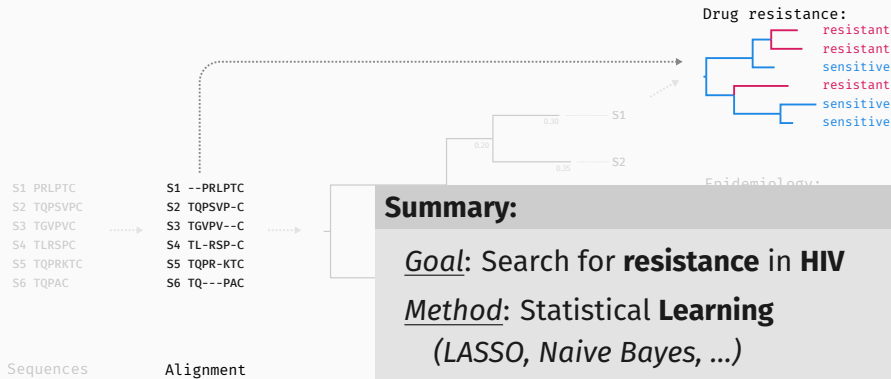
Blassel et al. 2022, RECOMB-SEQ; ¹Faure et al. 2025

keywords: Data Analysis, Pipelining, Machine Learning



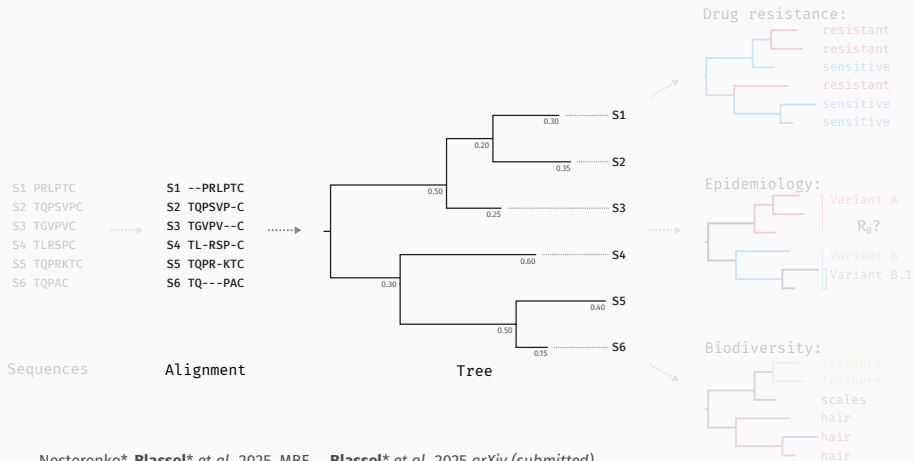
Blassel et al. 2021 PLoS CB & 2021 Curr. Op. in Vir

keywords: Data Analysis, Pipelining, Machine Learning

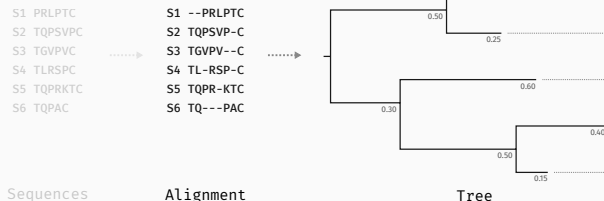


Blassel et al. 2021 PLoS CB & 2021 Curr. Op. in Vir

keywords: Pytorch, Deep Learning,



keywords: Pytorch, Deep Learning,



Summary:

Goal: Infer
tree distributions
from alignments

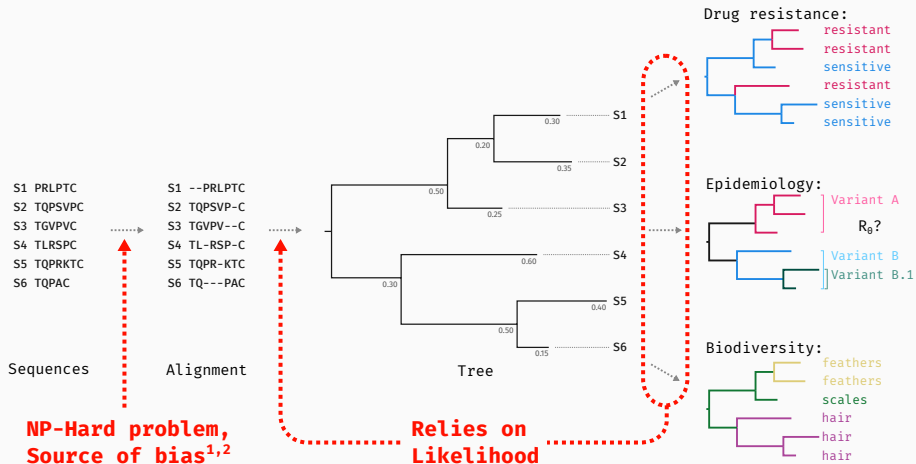
Method: Deep
neural networks
(Phyloformer 1&2)

Impact: Handles
intractable
models

Nesterenko*, Blassel* et al. 2025, MBE Blassel* et al. 2025 arXiv (submitted)

Context - Methodological Barriers

¹Liu et al. 2011, ²Wong et al. 2008



Context - The issue with likelihoods

State of the art relies on **Likelihood** ($\mathcal{L}$):

Max. Likelihood: $\theta^* = \underset{\theta}{\operatorname{argmax}} \mathbf{p}(\mathbf{x}|\theta, \mathbf{M})$

Bayesian: $p(\theta|x, M) \propto \mathbf{p}(\mathbf{x}|\theta, \mathbf{M})p(\theta)p(M)$

Costly^{1,2}

$$\mathcal{L} = \mathcal{O}(ns^3 + nLs^2)$$

Limits model-complexity

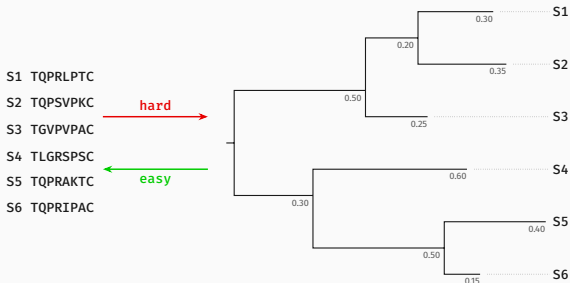
Settings with **impractical** $\mathcal{L}$

- **Potts:** models site **co-evolution**³
- **MTBD:** can model **epidemics** with distinct **populations**⁴

n : nb. of leaves, L : length of sequence, s : size of alphabet

¹Bryant et al. 2005, ²Felsenstein 2004, ³Rodriguez Horta et al. 2019, ⁴Barido-Sottani et al. 2020

Context - Likelihood-free inference

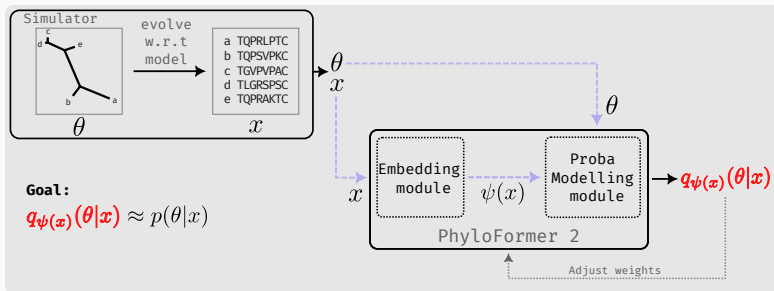


Setting:

- **Hard** to compute likelihood
- **Easy** to **simulate**

Solution:

1. **Simulate** many datapoints under model priors
2. **Learn** $f(x) : x \mapsto \theta$



Training: $\psi^* = \operatorname{argmax}_{\psi} \sum_{i \in \text{train}} q_{\psi(x_i)}(\theta_i|x_i)$ ¹

Inference: we **sample** from $q_{\psi^*(x_E)}(\theta_E|x_E)$

Blassel et al. 2025, *arXiv (submitted)*

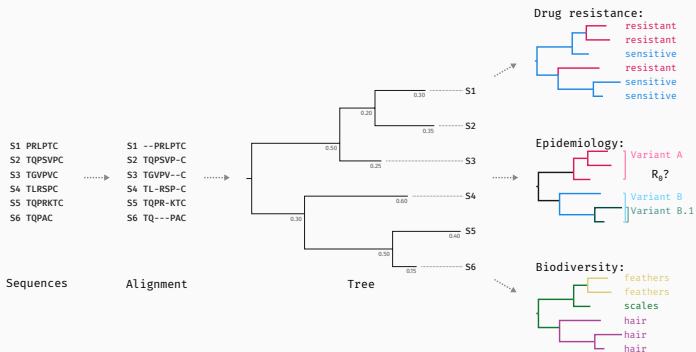
¹ equiv. to: $\operatorname{argmin}_{\psi} D_{\text{KL}}(q_{\psi(x)}(\theta|x) || p(\theta|x))$

Research Project

Project - Overview

Goal:

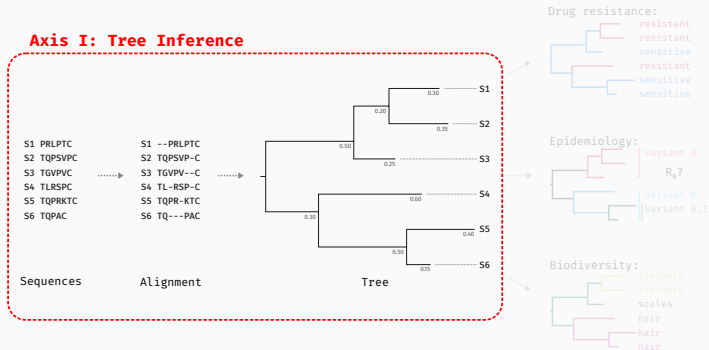
likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



Project - Overview

Goal:

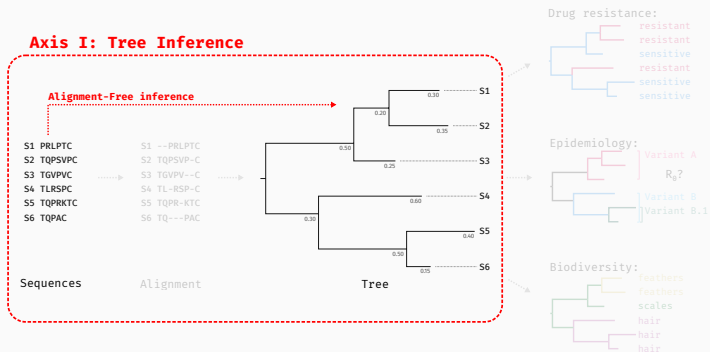
likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



Project - Overview

Goal:

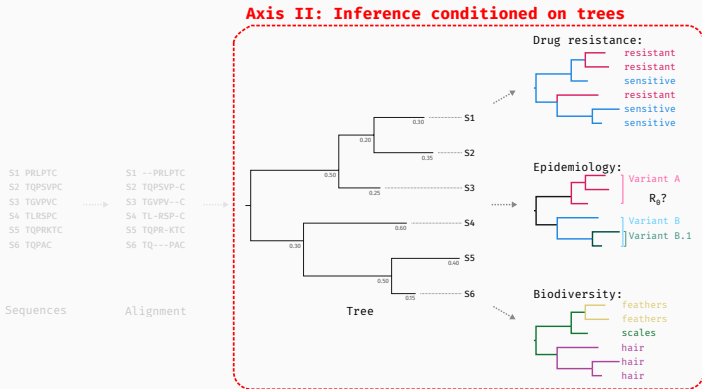
likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



Project - Overview

Goal:

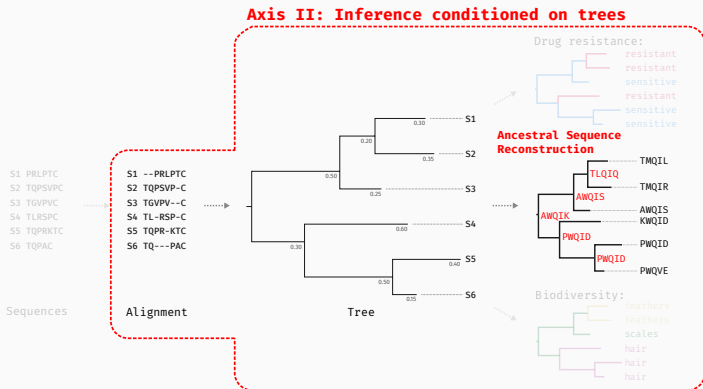
likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



Project - Overview

Goal:

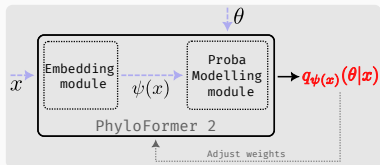
likelihood-free phylogenetic **methods** to enable the use of **richer, more realistic** models for phylogenetics



Project - Ax. I - Alignment and likelihood-free inference

Problem (Reminder):

- **Multiple** sequence **alignment** (MSA) is **hard**
- **Uncertainty** in MSA can be an important source of bias^{1,2}



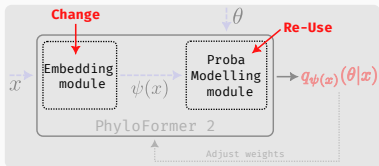
- **Hard** because we **lose** the **structure** induced by **MSAs**

¹Liu et al. 2011, ²Wong et al. 2008

Project - Ax. I - Alignment and likelihood-free inference

Problem (Reminder):

- **Multiple** sequence **alignment** (MSA) is **hard**
- **Uncertainty** in MSA can be an important source of bias^{1,2}



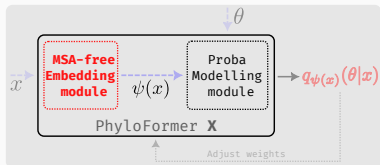
- **Hard** because we **lose** the **structure** induced by **MSAs**
- **Adapt** our existing posterior estimation **framework**

¹Liu et al. 2011, ²Wong et al. 2008

Project - Ax. I - Alignment and likelihood-free inference

Problem (Reminder):

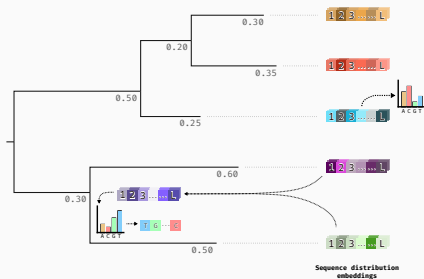
- **Multiple** sequence **alignment** (MSA) is **hard**
- **Uncertainty** in MSA can be an important source of bias^{1,2}



- **Hard** because we **lose** the **structure** induced by **MSAs**
- **Adapt** our existing posterior estimation **framework**
- **Not trivial** and this is only a **first-approach**

¹Liu et al. 2011, ²Wong et al. 2008

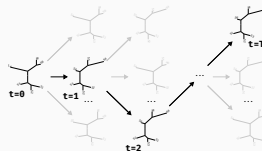
Project - Ax. II - Ancestral sequence reconstruction



- Representation of **sequence distribution** at each node
- *Training:* we have **simulated ancestral** sequences
- **Flexible** approach for other **branch-specific** parameters: Dates, speciation rates, ...

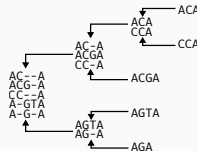
posterior calibration issues:

⇒ More **flexible** posterior modelling with **diffusion**



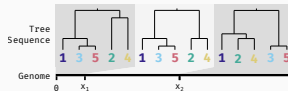
Alignments can still be **useful**:

⇒ **Integrate** alignment-free approach **to MSA** pipeline



Modelling **more complex** processes:

⇒ Infer **tree-sequences** to start tackling **recombination**



Scaling to large number of sequences:

- **17M** SARS-CoV-2 genomes in GISAID
- **900k** genomes in GTDB
- **250k** genomes in IGM database

Extend from **trees** to **networks**:

- **Horizontal transfer** is common in **bacteria**¹
- **Recombination** is frequent in **viruses**²
- *Collaboration with UW-Madison*

numbers: [GISAID](#), [GTDB](#), [IMG](#)

¹Arnold et al. 2022, ²Pérez-Losada et al. 2015

Integration & current collaborations

Analytical Genomics team @ CQSB:

Expertise: SBI/ML (L. Jacob), Evolution (M. Weigt, A. Carbone)

Collaborators: L. Jacob & P. Barrat-Charlaix, M. Weigt (*planned*)

Bioinfo team @ LISN:

Expertise: SBI/ML (F. Pouyet, F. Jay), Trees/Graphs (L. Planche)

Collaborators: F. Pouyet & F. Jay (*planned*)

Modelling of Pathogens team @ IBENS & IP:

Expertise: Evolution & Modelling (A. Zhukova, H. Morlon)

Collaborators: A. Zhukova

Research output:

6 Peer-reviewed **papers**

incl. 4 as (co)1st author

2 submitted **Preprints**.

Code on `git(hub|lab)`.



Supervision:

L3: 1, **M1:** 1, **M2:** 2

Oct 2026: **PhD** co-supervision
with **A. Zhukova**

Reviewing:

Journals: MBE, Syst. Bio, PLoS
CB, Bioinformatics

Conferences: RECOMB-SEQ,
ICLR, ICML

Collaborations:

France: CIRB, IBENS, Institut
Pasteur, LBBE

International: Penn. State,
EPFL, UW-Madison

Funded by:

PhD: INCEPTION & PRAIRIE

PostDoc: ANR DEELOGENY

**Thank you
for listening!**

lucblassel.com/CNRS_2026.pdf

References

- Arnold, B. J. et al. (2022). **Horizontal Gene Transfer and Adaptive Evolution in Bacteria.** In: *Nature Reviews Microbiology* 20.4, pp. 206–218.
- Barido-Sottani, J. et al. (2020). **A Multitype Birth–Death Model for Bayesian Inference of Lineage-Specific Birth and Death Rates.** In: *Syst Biol* 69.5, pp. 973–986.
- Bryant, D. et al. (2005). **Likelihood Calculation in Molecular Phylogenetics.** In: *Mathematics of Evolution and Phylogeny*. Ed. by O. Gascuel. Oxford University PressOxford, pp. 33–62.
- Faure, R. et al. (2025). **Alice: Fast and Haplotype-Aware Assembly of High-Fidelity Reads Based on MSR Sketching.** Pre-published.
- Felsenstein, J. (2004). **Inferring Phylogenies.**
- Liu, K. et al. (2011). **Multiple Sequence Alignment: A Major Challenge to Large-Scale Phylogenetics.** In: *PLoS Curr* 2, RRN1198.
- Pérez-Losada, M. et al. (2015). **Recombination in Viruses: Mechanisms, Methods of Study, and Evolutionary Consequences.** In: *Infect Genet Evol* 30, pp. 296–307.
- Rodriguez Horta, E. et al. (2019). **Toward Inferring Potts Models for Phylogenetically Correlated Sequence Data.** In: *Entropy* 21.11, p. 1090.
- Wong, K. M. et al. (2008). **Alignment Uncertainty and Genomic Analysis.** In: *Science* 319.5862, pp. 473–476.

Main slides

- Past: [Hist.](#), [Works](#)
- Future: [Ctxt.](#), [Proj.](#)
- Concl: [Int.](#), [Summ.](#)

Supp. Work

- PhD: [MSRs](#), [HIV DRMs](#)
- Phyloformers: [1](#), [2](#)
- PF2: [emb](#), prob
([tre](#),[mth](#),[ex](#),[samp](#))
- Production: [papers](#),[code](#)

Supp. Project

- [Overview](#)
- MSA-free: [1](#), [2](#), [3](#)
- ASM-free: [1](#)
- tree-seq: [1](#)
- diffusion: [1](#)
- GNNs: [1](#)
- Joint: [1](#)
- MSA guide: [1](#)

Published Articles:

- Nesterenko, L. et al. (2025). **Phyloformer: Fast, Accurate, and Versatile Phylogenetic Reconstruction with Deep Neural Networks**. In: *Molecular Biology and Evolution* 42.4, msaf051.
- Blassel, L., P. Medvedev, et al. (2022). **Mapping-Friendly Sequence Reductions: Going beyond Homopolymer Compression**. In: *iScience*, p. 105305.
- Blassel, L., A. Tostevin, et al. (2021). **Using Machine Learning and Big Data to Explore the Drug Resistance Landscape in HIV**. In: *PLOS Computational Biology* 17.8, e1008873.
- Blassel, L., A. Zhukova, et al. (2021). **Drug Resistance Mutations in HIV: New Bioinformatics Approaches and Challenges**. In: *Current Opinion in Virology* 51, pp. 56–64.
- Zhukova, A. et al. (2021). **Origin, Evolution and Global Spread of SARS-CoV-2**. In: *Comptes Rendus. Biologies* 344.1, pp. 57–75.
- Lemoine, F. et al. (2020). **COVID-Align: Accurate Online Alignment of hCoV-19 Genomes Using a Profile HMM**. In: *Bioinformatics* 37.12, pp. 1761–1762.

Submitted Preprints:

- Garot, V. et al. (2026). **Teddy: Neural Inference of Epidemiological Parameters from Viral Sequences**.
- Blassel, L., B. Boussau, et al. (2025). **Likelihood-Free Inference of Phylogenetic Tree Posterior Distributions**.

Pipelines

- DRMs (Snakemake): [github](#)
- MSRs (Nextflow): [github](#)
- PF data (Snakemake): [gitlab](#)

Contributions

- COVID-align (Go): [webserver](#)
- Tree CLI (Go): [github](#)
- MSA CLI (Go): [github](#)

Tools

- MSR tool (Go): [github](#)
- Phylo pkg (Rust+Python): [github](#)
- Tree-comp CLI (Rust): [github](#)
- PF 1 (Python+Pytorch): [github](#)
- PF 2 (Python+Pytorch): [gitlab](#)

Project - Global Project Overview

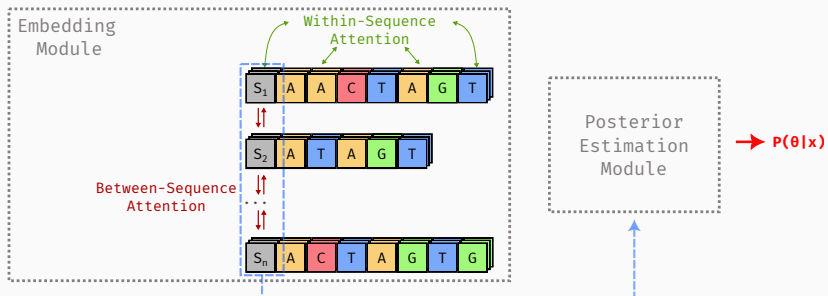
Axis 1: Likelihood-free end-to-end tree inference

- 1.1 [Alignment-free inference](#)
- 1.2 [Assembly-free inference](#)
- 1.3 [Tree-sequence inference](#)
- 1.4 [Tree-space diffusion](#)

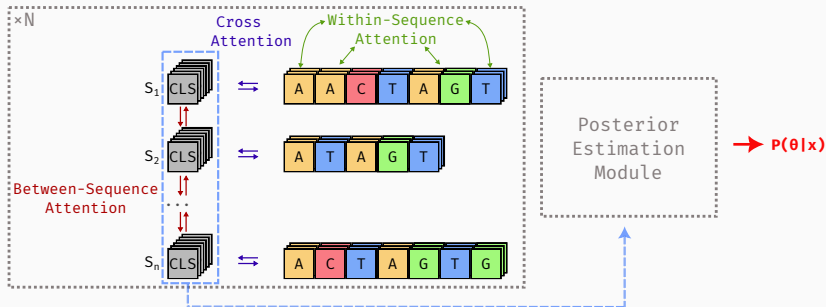
Axis 2: Neural inference conditioned on trees

- 2.1 [Neural inference on trees](#)
- 2.2 [Joint inference
trees+parameters](#)
- 2.3 [Neural inference
guided MSA](#)

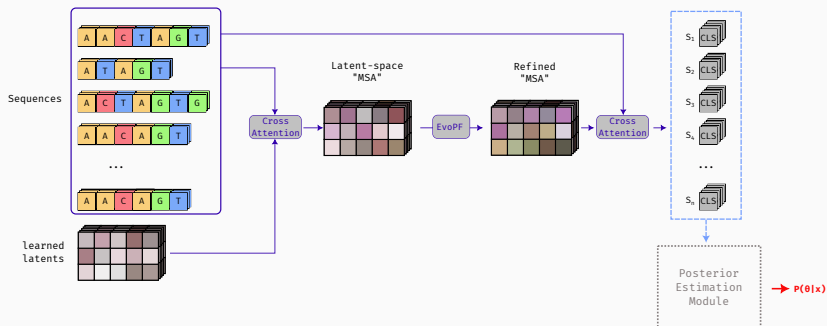
Project - Alignment-free embedding



Project - Alignment-free embedding



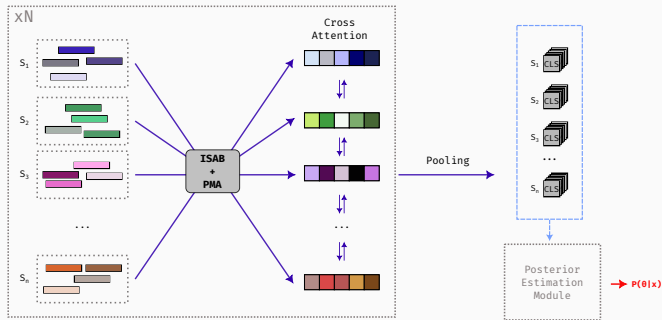
Project - Alignment-free embedding



Inspired by perceiver IO ¹

¹jaegle et al. 2022

Project - Assembly-free phylogenetic inference

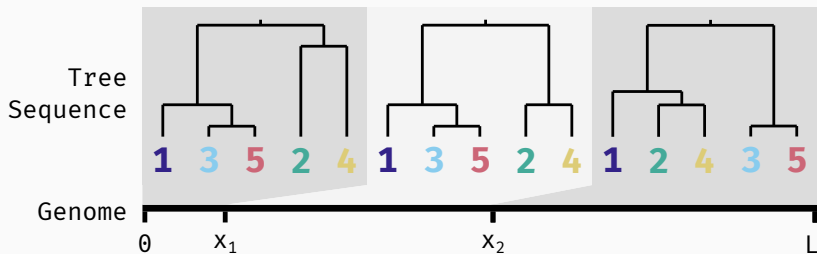


Project - Tree-sequence inference

Setting:

Recombination: sexual reproduction & viral co-infection.

Result: $\neq$ **regions** of the sequence evolve along $\neq$ **tree**.

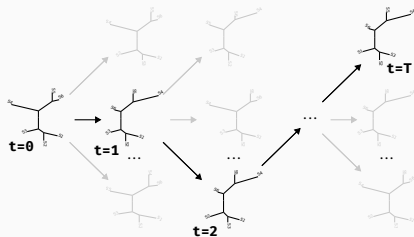


Approach:

1: **Add** a "sequence" **dimension** to embedding

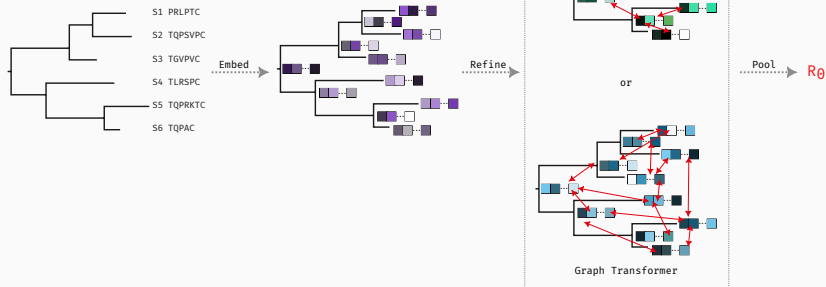
2: **Infer breakpoints** as well

Project - Discrete Diffusion in tree-space

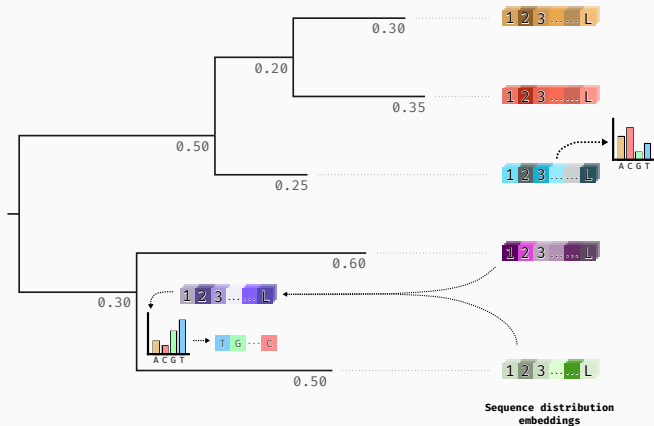


- Denoising diffusion model
- Noising process:
 - topological moves (1st NNIs, then SPRs)
 - noise applied to categorical distrib
- Enforces tree-structure at every step

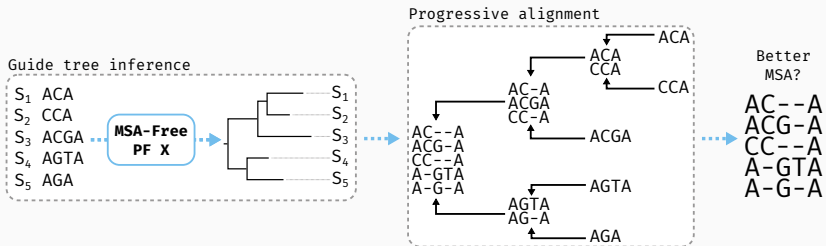
Project - Inference on trees, GNNs for R_0



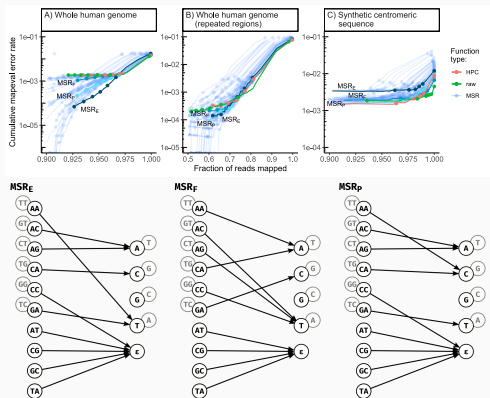
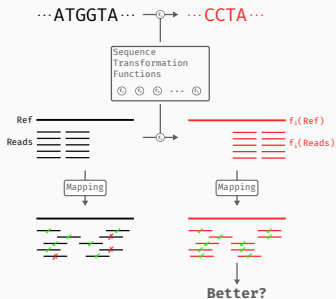
Project - Joint tree-parameter inference, ASR



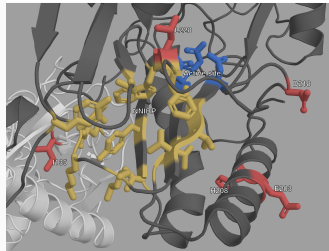
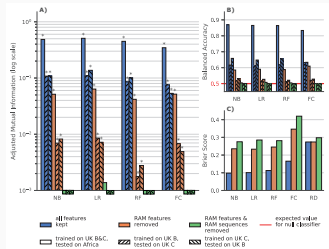
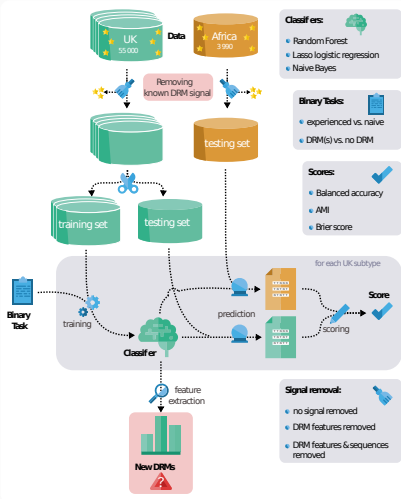
Project - Alignment-free tree guided MSA

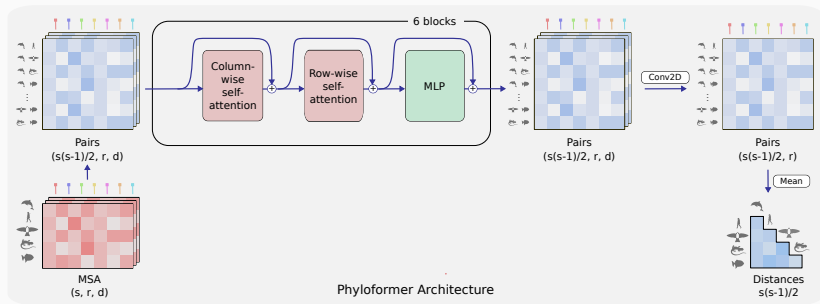


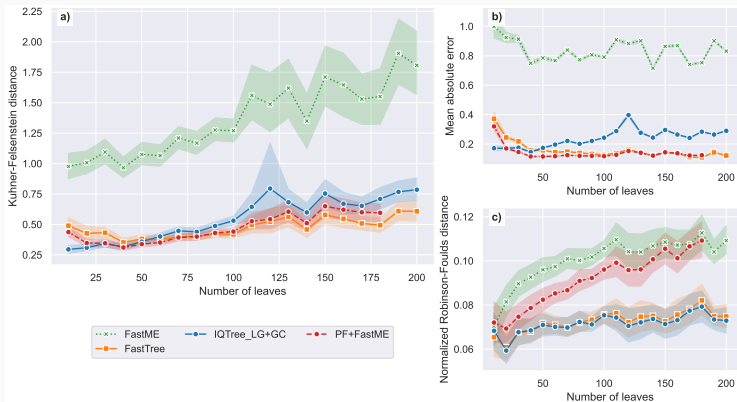
Previous work - Mapping-friendly Sequence reductions



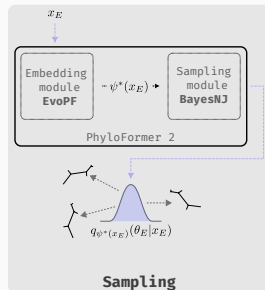
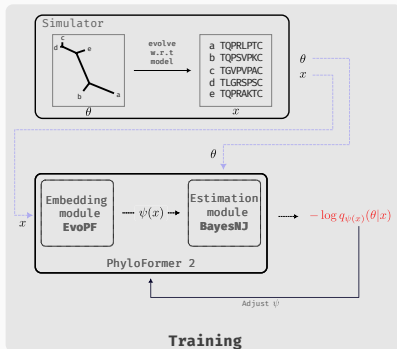
Previous work - Finding DRMs in HIV



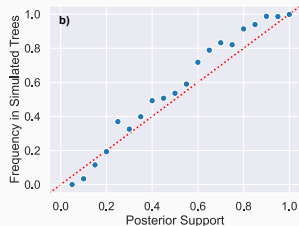
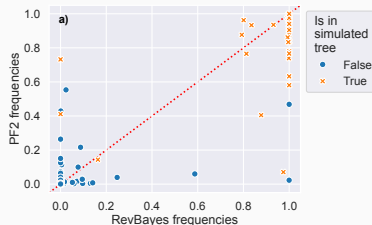
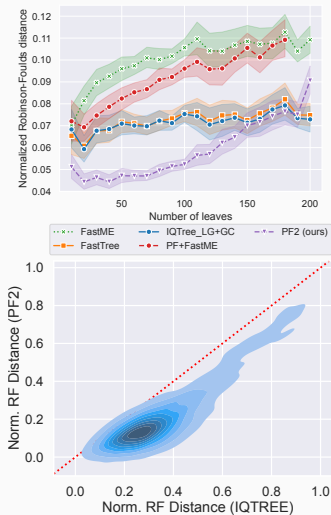


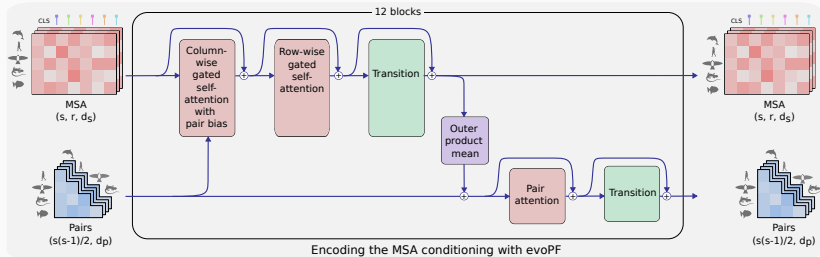


Previous work - Phyloformer 2

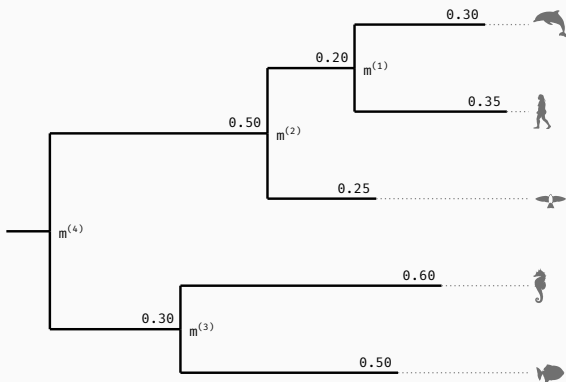


Previous work - Phyloformer 2 (results)

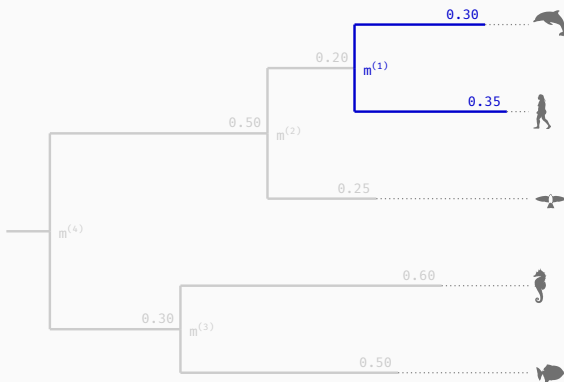




We want to describe the following tree:

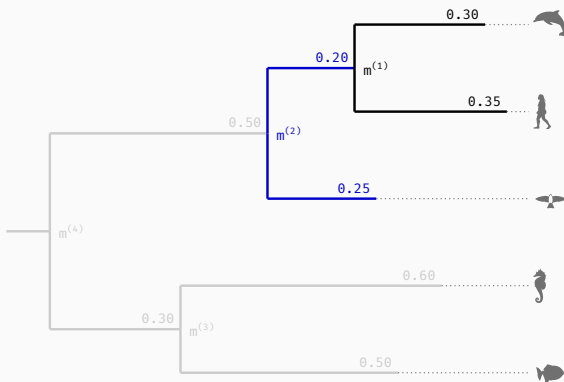


Iteratively merge shortest cherry:



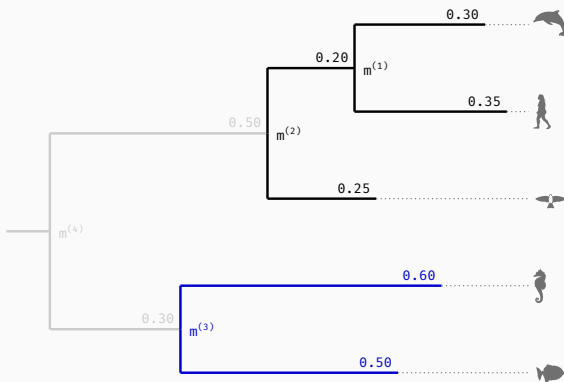
$$\tau = \{m^{(1)}\}$$

Iteratively merge shortest cherry:



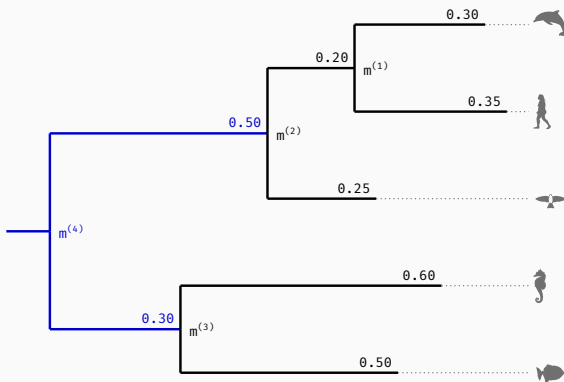
$$\tau = \{m^{(1)}, m^{(2)}\}$$

Iteratively merge shortest cherry:



$$\tau = \{m^{(1)}, m^{(2)}, m^{(3)}\}$$

Iteratively merge shortest cherry:



$$\tau = \{m^{(1)}, m^{(2)}, m^{(3)}, m^{(4)}\}$$

- **Tree** is an **ordered set** of merges: $\theta : \{m^{(1)}, \dots, m^{(N-1)}\}$
- We **factorize** $q_{\psi(x)}(\theta|x)$ as the product of successive merge probabilities:

$$q_{\psi(x)}(\theta|x) = \prod_{k=1}^{N-1} q_m(m^{(k)}|m^{(<k)}) q_\ell(\ell^{(k)}|m^{(\leq k)})$$

- **Merge probabilities have 2 components:**
 - topological:** $q_m(m^{(k)}|m^{(<k)})$
 - branch-length:** $q_\ell(\ell^{(k)}|m^{(\leq k)})$

